Simple Case of Graham and Powell (2012)

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1 one unit

Start with one unit and an explicit model. For this single i, suppose

$$Y_{it} - W_{it}\delta_0 = X_{it}\beta_i + \varepsilon_{it}, \qquad t = 1, \dots, T,$$

where β_i is fixed and δ_0 is known. Use the example

$$X_{it} = S_i Z_{it}, \qquad S_i \sim U[0, b], \qquad Z_{it} \overset{\text{iid}}{\sim} N(0, 1).$$

Let the errors be independent of the regressors and have dentity cov (as in example) also for now assume normality. Creates simple classic OLS case.

Regression is

$$\boxed{b_i(X) - \beta_i = \frac{\sum_{t=1}^{T} X_{it}\varepsilon_{it}}{\sum_{t=1}^{T} X_{it}^2}.}$$

Now calculate its conditional squared error explicitly:

$$\begin{aligned} E[(b_i(X) - \beta_i)^2 \mid X_i] &= \frac{E\left[\left(\sum_t X_{it}\varepsilon_{it}\right)^2 \mid X_i\right]}{\left(\sum_t X_{it}^2\right)^2} \\ &= \frac{\sum_t \sum_s X_{it}X_{is}E(\varepsilon_{it}\varepsilon_{is} \mid X_i)}{\left(\sum_t X_{it}^2\right)^2}. \end{aligned}$$

Thus,

$$\boxed{\text{Given the observed } X_i, \quad \operatorname{Var}(b_i(X_i) \mid X_i) = (X_i'X_i)^{-1}.}$$

Because the same S_i multiplies every period,

$$\begin{aligned} \frac{1}{T}X_i'X_i = S_i^2 \frac{1}{T}\sum_t Z_{it}^2 \\ \longrightarrow S_i^2 \quad \text{almost surely.} \end{aligned}$$

More periods average the Z_{it}^2. They do not average the persistent S_i^2. Consequently,

$$\left(\frac{1}{T}X_i'X_i\right)^{-1} \longrightarrow \frac{1}{S_i^2},$$

a random limit across possible unit scales. For each fixed $s^2 > 0$,

$$\boxed{\sqrt{T}\,[b_i(X) - \beta_i] \mid S_i^2 = s^2 \xrightarrow{d} N(0, 1/s^2).}$$

For the unconditional single-unit limit, write $G \sim N(0,1)$, independent of S_i. Integrating the conditional distributions gives

$$\boxed{\sqrt{T}[b_i(X_i) - \beta_i] \xrightarrow{d} \frac{G}{S_i}.}$$

$$E\left[\left(\frac{G}{S_i}\right)^2\right] = E[G^2]E[S_i^{-2}] = \frac{1}{b}\int_0^b \frac{ds}{s^2} = \infty.$$

So the single-unit limiting mixture has infinite second moment. Also,

$$E\left[\left|\frac{G}{S_i}\right|\right] = \frac{\sqrt{2/\pi}}{b}\int_0^b \frac{ds}{s} = \infty.$$

Mean does not exist either.

2 N units

Moving from a fixed unit to a growing number of units.

This creates N normal observations with respective constant variances at $\sqrt{T}$ scaling. Suppose N=T both growing. Then $\sqrt{N}$ estimator will mix these normal distributions. Its second moment is

$$\frac{1}{b}\int_0^b \frac{ds}{s^2} = \infty.$$

or more explicitly

Independence and normality give

$$\boxed{\sqrt{N}\left[\frac{1}{N}\sum_i b_i(X_i) - \frac{1}{N}\sum_i \beta_i\right] \;\Big|\; X_1, \ldots, X_N \sim N\left(0, \frac{1}{N}\sum_i \frac{1}{X_i' X_i}\right).}$$

For every N, T. When $T = N$, its conditional variance is

$$\boxed{\frac{1}{N^2}\sum_{i=1}^N \frac{1}{S_i^2\left(N^{-1}\sum_{t=1}^N Z_{it}^2\right)}.}$$

Thus the unconditional distribution is a normal mixture. Its second moment is already infinite:

$$E\left[\left\{\sqrt{N}\left[\frac{1}{N}\sum_i b_i(X_i) - \frac{1}{N}\sum_i \beta_i\right]\right\}^2\right] = E\left[\frac{1}{N}\sum_i \frac{1}{X_i' X_i}\right] = \infty.$$